

# CSE2315 Lab Course

## Assignment 7

- This assignment consists of several exam-level questions and sometimes an optional extra exercise.
- See Brightspace, *Course Information* for instructions about the lab course, rules and grading procedure.
- Your solution has to be handed in on paper, typeset in  $\text{\LaTeX}$ , using a word processor or readable handwriting. Handing in via email is not permitted.
- Cooperating with a colleague is permitted and *strongly encouraged!* In such cases, please hand in a single copy of the work.
- Total number of pages, without this cover page: 2.

1. Prove that the following problem is in NP:

$\{(V, E) \mid \text{The graph } G = (V, E) \text{ contains a simple path } \pi \text{ such that } \forall v \in V : v \in \pi\}$

In a different notation we would write:

**Name:** HAMILTONIAN CYCLE (HAMC)

**Input:** A graph  $G = (V, E)$ .

**Question:** Is there a simple cycle  $\pi$  in  $G$  such that all vertices  $v \in V$  are in that cycle?

2. *Old exam question, endterm 23-24:*

For the following claim, if it is true: prove it. If they are false, give an explanation showing the claim is false and if possible a counterexample. Remember that the cover of our exam reads: "Unless otherwise stated we assume that  $P \neq NP$  and that  $NP \neq EXP$ ."

For three languages  $A, B, C$  if  $A \leq_P B$  and  $B \leq_m C$  with  $C \in NPC$ . Furthermore we know that there is a regular expression  $R$  such that  $L(R) = A$ . Then  $B \in P$ .

3. (Old exam question) On the (famous?) British Quiz Show "The University Challenge", the following question was asked in the grand final of 2018/2019.

[...] which problem in computing asks whether all problems with a solution verifiable in finite time, can also be solved in finite time.

The contestants answer with "P equals NP" and the host answers back with "Yes that's right, P versus NP, yes".

Assuming this is the answer the host was looking for, what is wrong with the question he asks?

4. *Old exam question, resit 20-21:*

- (a) Prove that the following problem is in NP.

**Name** AMAZINGCUSTOMCHALLENGE

**Instance** Given a set  $L$  of lectures, a set of students  $S$ , a collection of subsets  $T$  with  $|T| = |S|$  so that  $T_i$  is the set of lectures that student  $i$  needs to attend, and an integer  $K$ .

**Question** Is there a function  $f : L \rightarrow \{1, 2, \dots, K\}$  that maps lectures to time slots so that all students can attend all their lectures. In other words is there a function so that there is no student who has to attend two lectures that are mapped to the same time slot.

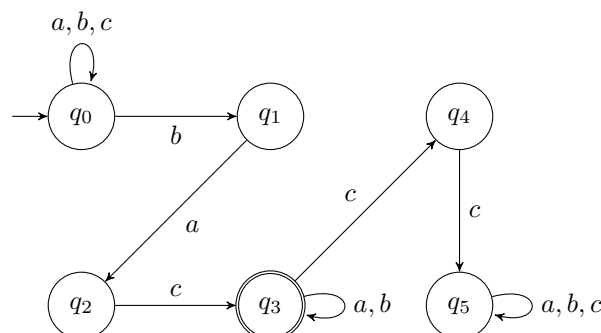
- (b) *Not originally part of the exam question* Your answer to the previous question should have included time complexities. If we are to describe them in terms of  $|I|$  where  $I$  is an instance of the problem, what would be the complexity? (Hint: first think about  $|I|$ , is it  $|S|$ , or  $|L|$ , or some combination? How does  $K$  factor into this?)

5. *Variation on old exam question, resit 20-21:*

Show the claim is true or false, starting your answer with either "True" or "False".

Given two problems  $A$  and  $B$  that can be solved using a PDA, it is possible to create a Karp-reduction from  $A$  to  $B$  and vice versa.

6. *Revision, old exam quesiton, resit 20-21:* Given this NFA  $N$ , give a regular expression  $R$  so that  $L(N) = L(R)$ .



7. *Revision, old exam question, endterm 16-17:* Suppose we have the following language:

$$L = \{\langle M, v, w \rangle \mid M \text{ outputs } w \text{ on the tape when run on input } v\}$$

Use a mapping reduction to prove that  $L$  is undecidable. Give:

- (a) A suitable problem to reduce from.
- (b) A suitable reduction function  $f : \Sigma^* \rightarrow \Sigma^*$ .
- (c) A proof showing  $f$  satisfies the requirements of a mapping reduction.